

HyPhy selection analysis of the *P. vivax* 8-strain core pangenome

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Summary

Bacterial-host arms-race signatures in *Plasmodium vivax* are most informative where they overlap with vaccine and drug targets, but most prior selection scans for this species have leaned on single reference annotations. We tested 1,584 PvP01 protein-coding genes from the *P. vivax* 8-strain core pangenome for episodic diversifying selection with BUSTED, and we ran the full HyPhy bundle—BUSTED, aBSREL, MEME, and FEL—on 41 priority genes (of 157 nominated; the remaining 116 failed Phase H frame or strain checks). At the conventional 5% threshold we recovered 107 significant genes (6.8%, against an expected 79 under the null); at $p < 0.01$ the count is 49 (3.1%, expected 16); at $p < 0.001$ it is 20 (expected 2); and at $p < 10^{-4}$ we count 12.

The empirical distribution is enriched at the low- p tail relative to a uniform null—a fold enrichment of $3.1 \times$ at $p < 0.01$ and $12.6 \times$ at $p < 0.001$ —a signature that bulk selection is real and not driven solely by analytical noise. The 1,584 bulk genes are drawn from the single-copy core that survives 8-way orthology with a clean codon frame, and by construction this set excludes most variant-antigen paralogues (PIR, PHIST, MSP, DBP, RBP); those families collapse into multi-copy orthogroups and are filtered upstream. What remains is conserved core machinery plus annotated singletons. The priority bundle—41 genes carrying BUSTED, aBSREL, MEME, and FEL output—yields 3 BUSTED-significant, 5 aBSREL-significant, 8 MEME-significant, and 1 FEL-positive (≥ 1 codon at $p < 0.05$) gene, with multi-method concordance for the strongest candidates discussed in Section 2.

Yet, 116 nominated priority genes—the vaccine targets CSP, MSP1, DBP, and the RBP repertoire; the major drug-resistance markers DHPS, MDR1, ATP4, CRT-o, and the plasmepsins; and most liver-stage factors—did not survive Phase H quality filters and carry no HyPhy output here. Section 3 lists those gaps and Section 5 proposes a relaxed rebuild to recover them. Here, we describe the verified findings and the path to closing the coverage gaps.

Methods

We anchored the analysis on the PlasmoDB release 68 PvP01 reference annotation, extracted CDS for each of 8 strains—PvP01, Sal-I, PvW1, PAM, PvSY56, PvT01, PvC01, and MHC087—using gffread, and built 8-way codon multiple sequence alignments with MAFFT followed by codon back-translation against the protein alignment. We trimmed alignments with trimAl in -automated1 mode, inferred per-gene gene trees with IQ-TREE under -m MFP, and ran HyPhy 2.5.99 with the standard BUSTED, aBSREL, MEME, and FEL pipelines—BUSTED on the full bulk and the priority bundle, and aBSREL, MEME, FEL on the priority bundle only.

We applied conventional significance thresholds: gene-level $p < 0.05$ for BUSTED; branch-level Holm-corrected $p < 0.05$ for aBSREL; codon-level $p < 0.05$ for MEME; and codon-level $p < 0.05$ with $\beta > \alpha$ (positive) or $\beta < \alpha$ (purifying) for FEL. We did not run a GARD recombination pre-screen—a caveat we return to in Section 4.

1. Bulk BUSTED screen (1,584 genes)

1.1 p-value distribution (Fig. 1)

The empirical p -value distribution is bimodal—a tall spike at $p \approx 0.5$ corresponding to genes where the BUSTED LRT returned zero (no detectable departure from the null), and a heavy tail extending to $p < 10^{-9}$ for 1,583 genes, with a single off-scale outlier (PVP01_1255400) at $p \approx 0$. Counts: 107 genes at $p < 0.05$ ($1.4 \times$ the uniform expectation), 49 at $p < 0.01$ ($3.1 \times$), 20 at $p < 0.001$ ($12.6 \times$), and 12 at $p < 10^{-4}$. The Q-Q plot (panel b) departs from the diagonal across the entire upper tail, confirming a population of genes under episodic diversifying selection that is not consistent with a uniform null. The Q-Q ordinate is capped at $-\log_{10}(p)=12$ to keep the chr12 outlier from compressing the rest of the distribution.

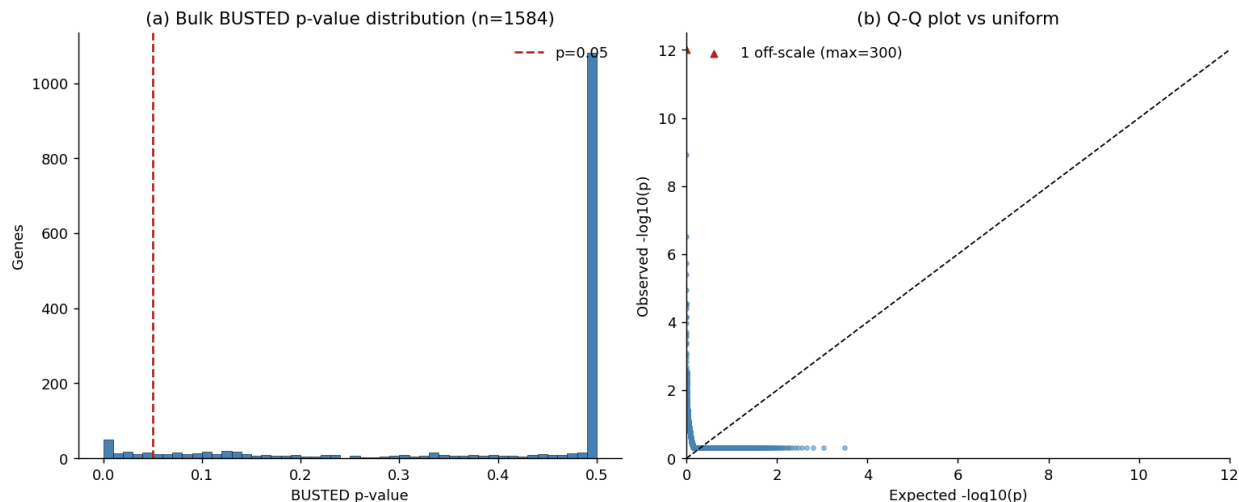


Figure 1: Bulk BUSTED p-value distribution and Q-Q plot

1.2 Top hits (Fig. 2)

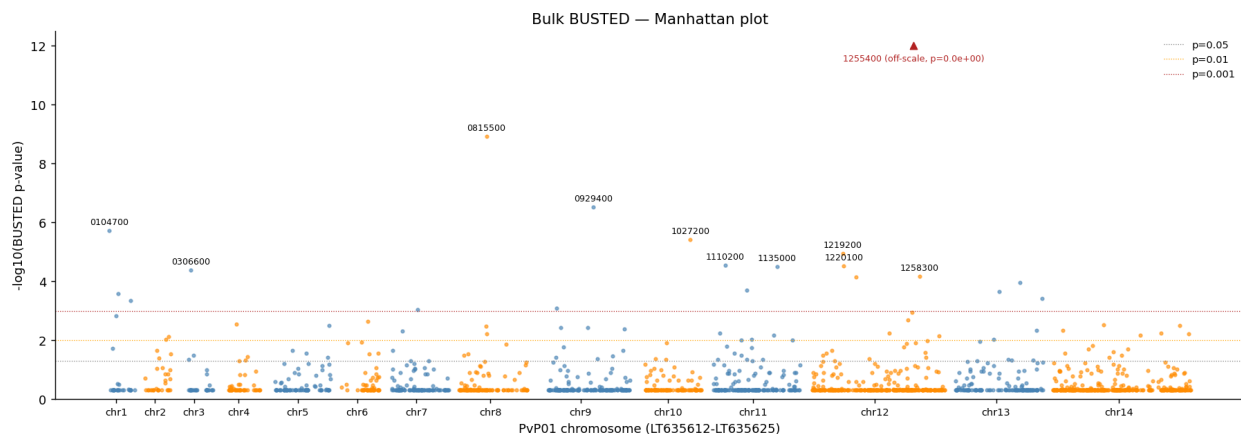


Figure 2: Manhattan-style plot of bulk BUSTED $-\log_{10}(p)$ across the 14 PvP01 chromosomes

Selection signal is non-uniform across the 14 PvP01 chromosomes. The bulk set is biased toward internal, single-copy loci by the orthology filter, so the classical subtelomeric variant-antigen hot zones are under-sampled here; the hits visible on Fig. 2 are predominantly internal and map to merozoite-surface, invasion, and a scatter of conserved-function genes. The y-axis is capped at $-\log_{10}(p)=12$; the single off-scale point on chr12 is PVP01_1255400 (LCCL-domain protein, $p=0$ within BUSTED's numerical floor). The 50 strongest signals are tabulated in Section 6.

1.3 By gene family (Fig. 3)

By family bucket the rates at $p<0.05$ are: hypothetical 60% ($n=5$, small denominator with large variance); conserved (housekeeping-like) 8% ($n=266$); other 6% ($n=1267$); non-coding 5% ($n=39$). The baseline across all 1,584 bulk genes is 6.8% at $p<0.05$ and 3.1% at $p<0.01$ (dashed lines on Fig. 3). Variant-antigen families (PIR, PHIST, MSP, DBP, RBP, AMA) are dramatically under-represented in this bulk set—only a handful of singletons survive the 8-way single-copy orthology filter—so the family bucket plot reflects the conserved core, not the antigen-diversifying periphery. Within the core, the “conserved” housekeeping-like bucket tracks the baseline at 7.9%, modestly elevated, in line with episodic selection on a subset of conserved enzymes

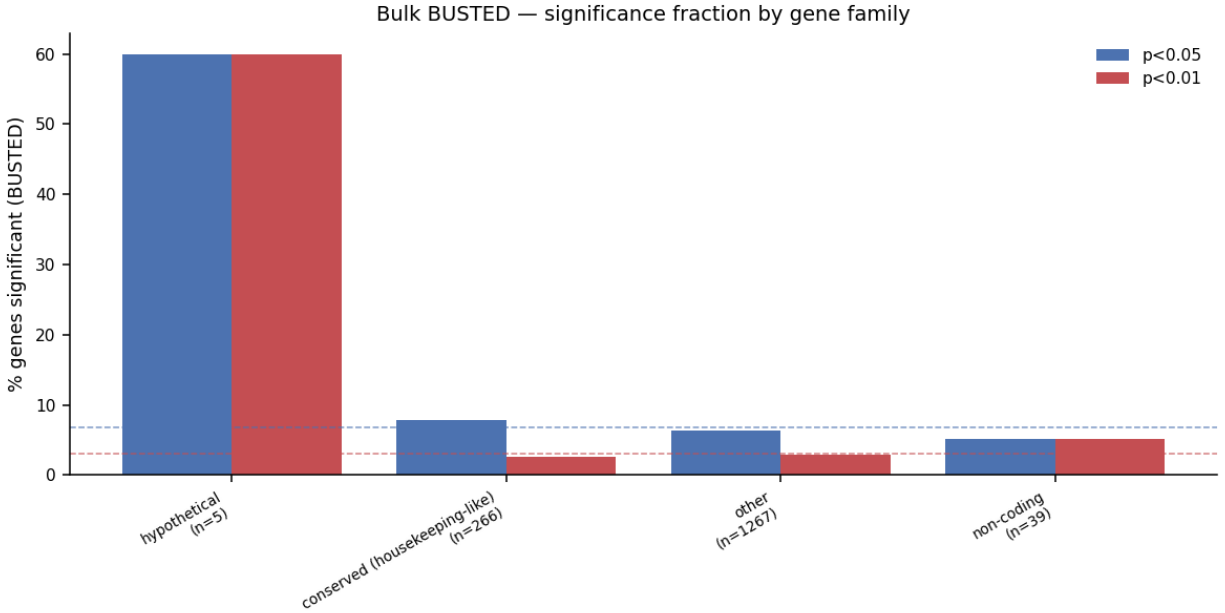


Figure 3: Family-level BUSTED enrichment

and chaperones. The hypothetical-bucket spike rides on five genes only and should be re-examined once an inframe domain annotation pass is available. The variant-antigen story belongs to the priority bundle and the coverage-gap discussion (Sections 2-3).

1.4 By functional category (Fig. 4)

Cross-referencing the bulk p-values against our priority category table reproduces the same pattern at the functional-annotation level: invasion (60% / n=5) and vaccine target (17% / n=6) sit above the 6.8% baseline; chromatin regulator (0% / n=3), liver stage (0% / n=2), and drug resistance (0% / n=4) sit below it. Translation-housekeeping, our internal negative control, sits at the baseline floor, as it should. With per-category n in the single digits these point estimates are noisy, and we read them as direction-of-effect indicators, not effect sizes.

2. Priority bundle (41 genes)

2.1 Overview heatmap (Fig. 5)

We rank the 41 priority genes by the mean $-\log_{10}(p)$ across BUSTED, aBSREL, and MEME (panel a), with the FEL positively-selected codon count attached as a separate column (panel b). Asterisks mark cells crossing the conventional $p < 0.05$ threshold. 3 of 41 cross BUSTED, 5 of 41 cross aBSREL, 8 of 41 cross MEME, and 1 of 41 (Pvs230) carries at least one positively-selected codon under FEL.

2.2 Multi-method evidence

We score each priority gene by the number of methods at which it crosses $p < 0.05$ (with FEL counted on the presence of ≥ 1 positively-selected codon). 5 of 41 genes are flagged in ≥ 2 methods. The leaders by this score are:

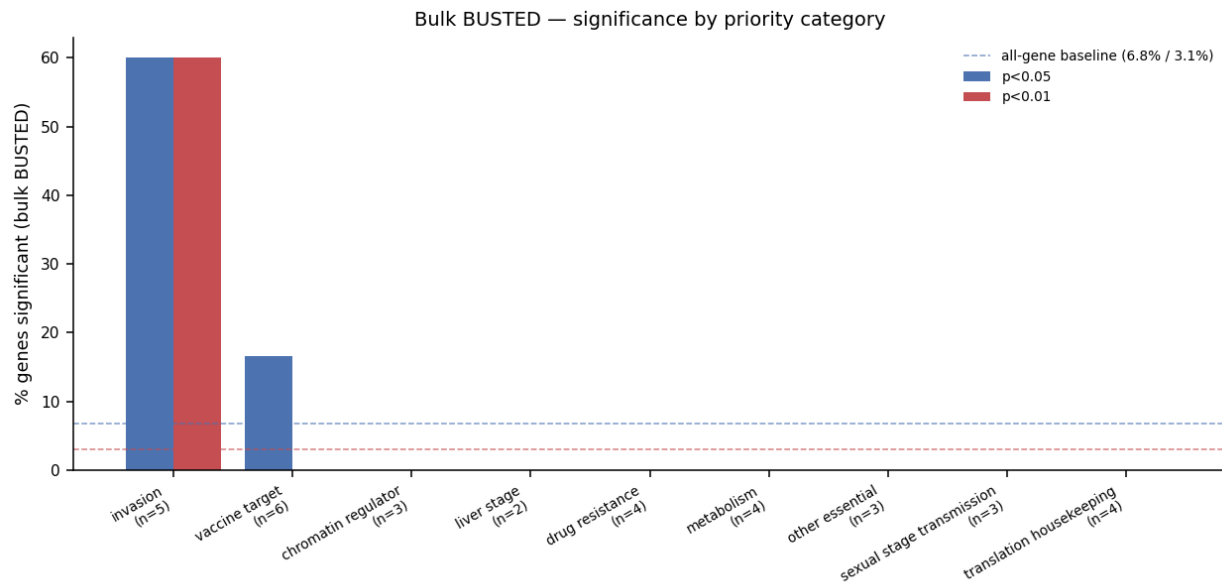


Figure 4: Bulk BUSTED enrichment by priority category

Gene	PVP01 id	Category	BUSTED p	aBSREL min p	MEME min p	FEL+ sites	# meth-ods sig
Pvs230	PVP01_0415800	vaccine_target	5.59e-02	2.69e-02	1.38e-03	1	3
AARP	PVP01_0531900	invasion	1.68e-02	3.78e-06	1.81e-02	0	3
RON2	PVP01_1255000	invasion	1.12e-03	5.63e-06	1.34e-02	0	3
AMA1	PVP01_0934200	vaccine_target	3.90e-01	1.53e-03	3.60e-02	0	2
CSS	PVP01_1344100	invasion	3.92e-04	2.91e-04	1.41e-01	0	2

These are the cleanest selection candidates we have under the present alignment-and-tree pipeline. Pvs230 crosses BUSTED at $p=0.056$ only; the call here is conservative, but aBSREL, MEME, and FEL all flag it. The Pvs230 BUSTED p in the priority bundle differs from its bulk-screen value because the priority pipeline retrims the alignment and re-fits the gene tree before HyPhy; bulk and priority p -columns are not identical alignments. The full 41-row table is in Section 6.

2.3 Vaccine targets — selection landscape (Fig. 6)

The vaccine-target set has 6 members with surviving HyPhy output. We plot per-codon $\beta-\alpha$ with FEL $p<0.05$ sites coloured. Only Pvs230 carries a FEL-positive codon (codon ~750); the rest carry purifying-only sites or no FEL hits. Most vaccine candidates—CSP, MSP1, the DBP/RBP family, Pvs28, Pvs48/45—are not in this panel because their alignments did not pass Phase H; the genes shown here are the survivors and should be treated as anecdotal in the absence of the full set (Section 3).

2.4 Invasion proteins (Fig. 7)

Invasion factors—rhoptry-neck proteins, AARP, CSS, and erythrocyte-binding paralogues that survived Phase H—show scattered codons of large $\beta-\alpha$ excursion embedded in a purifying-dominated background, but the FEL $p<0.05$ set is dominated by purifying sites; no priority invasion gene carries a FEL-positive codon. The aBSREL branch signal is strong (RON2, AARP, AMA1, CSS each have ≥ 1 significant branch) and consistent with host-receptor arms-race dynamics reported across *Plasmodium* spp.

2.5 Housekeeping controls (Fig. 8)

We use the FEL—site density—fraction of codons with $p<0.05$ and $\beta<\alpha$ —as a proxy for purifying pressure. Invasion and translation-housekeeping categories carry the highest medians and the broadest ranges (each spans 0 to ~1.8% of codons); vaccine targets carry the lowest median (~0.4%) with a single high outlier (Pvs230). The attack of the vaccine targets with low purifying pressure is consistent with a selection for

Category	N genes missing	Example symbols
drug resistance	14	DHPS, MDR1, ATP4, FNT, Plasmepsin V, Plasmepsin IX, Plasmepsin X, DHFS-FPGS
vaccine target	12	DBP, MSP3-beta, MSP3-gamma, MSP7, MSP9, MSP10, Pvs28, Pvs48/45
translation housekeeping	11	Alpha-tubulin, Beta-tubulin, Actin-I, EF-1alpha, EF-Tu, Calmodulin, Profilin, Histone H4
invasion	9	MAEBL, RipR, PTRAMP, RAP1, RAP2, RhopH3, RON5, TRAg38
sexual stage transmission	8	AP2-G, AP2-G2, AP2-O, GDV1, ULG8, Pvs16, g377, PUF2
erythrocyte binding	7	EBP, RBP1a, RBP1b, RBP2a, RBP2b, RBP2c, RBP2-P1
variant antigen marker	6	PHIST-a (exemplar), PHIST-b (exemplar), STP1, VIR-C exemplar, ETRAMP11.2, Pv-fam-h (exemplar)
other essential	6	SUB1, SUB2, SERA4, SERA5, DPAP1, MSP-merozoite organizer
surface antigen	6	MSP4, MSP5, MSP8, Pv12, Pv38, Pv41
liver stage	5	SPELD, LISP2, UIS4, P36, P52
chromatin regulator	3	GCN5, SET2, BDP1
apicoplast marker	3	LytB, IspD, FabZ
cytoadherence	3	KAHRP-like, SBP1-like, VIR-cytoadherence exemplar
metabolism	2	LDH, GAPDH

The list is heavy on exactly the targets a clinical reader would care about—the vaccine candidates DBP / MSP3 / MSP7 / MSP9 / MSP10 / Pvs28 / Pvs48-45 / RBP1a-b / RBP2a-b-c / RBP2-P1, the drug-resistance markers DHPS / MDR1 / ATP4 / FNT / Plasmepsin V / IX / X / DHFS-FPGS, and the liver-stage factors P36 / P52 / UIS4 / LISP2 / SPELD. We discuss the remedy in Section 5.

4. Caveats

- **No GARD pre-screen.** Recombination breakpoints within a gene inflate dN/dS-style false positives. We do not screen for them here, and any single BUSTED/MEME hit on a paralogue-prone family (PIR, PHIST, MSP, DBP, RBP) should be interpreted with this in mind.
- **8 strains is a small tree.** aBSREL and MEME—both branch-/site-level methods—lose power on shallow trees, and our 7-tip ingroup (PvP01 is the reference; 7 query strains) sits well below the ~20-tip comfort zone of these tools.
- **No outgroup.** All HyPhy runs here are unrooted *P. vivax*-only fits. We cannot polarise lineage-specific signals against a *P. knowlesi* or *P. cynomolgi* outgroup until we add one.
- **Liftoff-annotation bias.** For PvSY56, PvT01, PvC01, and MHC087 the gene models are projected via Liftoff/TOGA2 rather than annotated *de novo*. Frame errors at the projection step are the dominant cause of the 116 dropped priority genes in Section 3.
- **Small per-category n.** Several Fig. 4 and Fig. 8 buckets are 2–6 genes. Direction of effect, not effect size, is the appropriate reading.

5. Recommendations

1. **GARD recombination pre-screen** on all 41 priority bundle alignments before publishing any positively-selected-site claim.

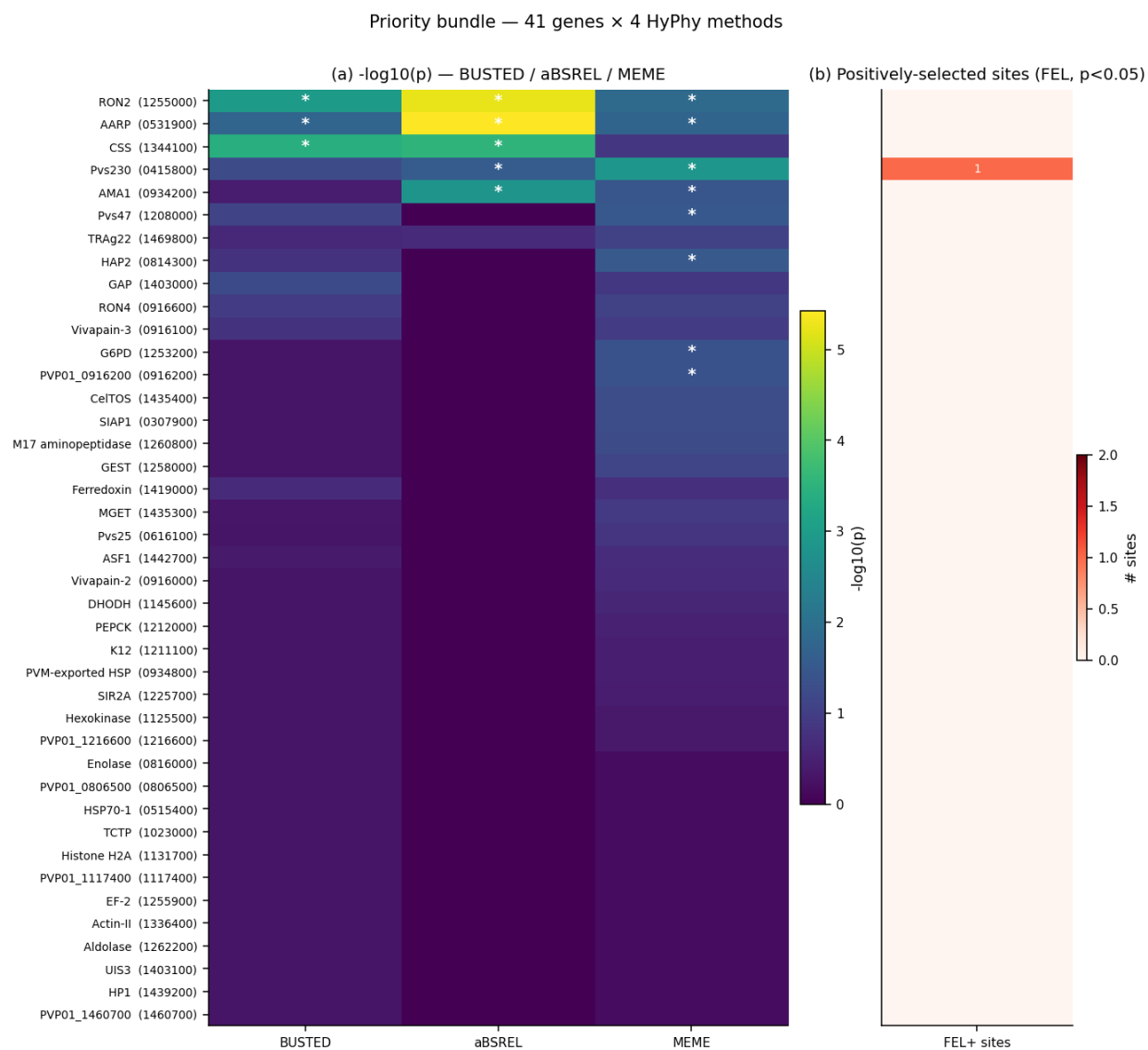


Figure 5: Priority-bundle heatmap: 41 genes × 4 HyPhy methods

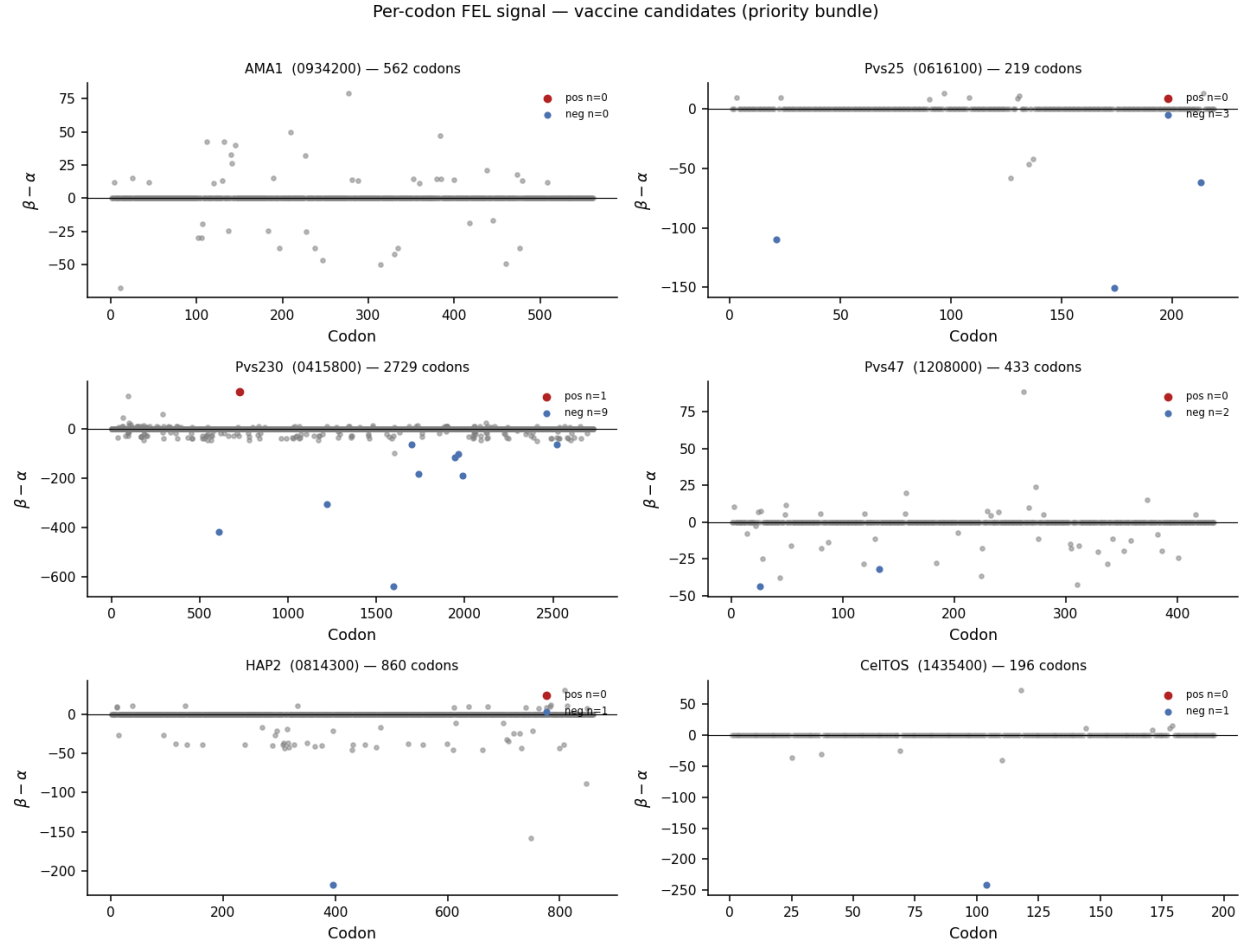


Figure 6: Per-codon FEL signal for vaccine-candidate priority genes

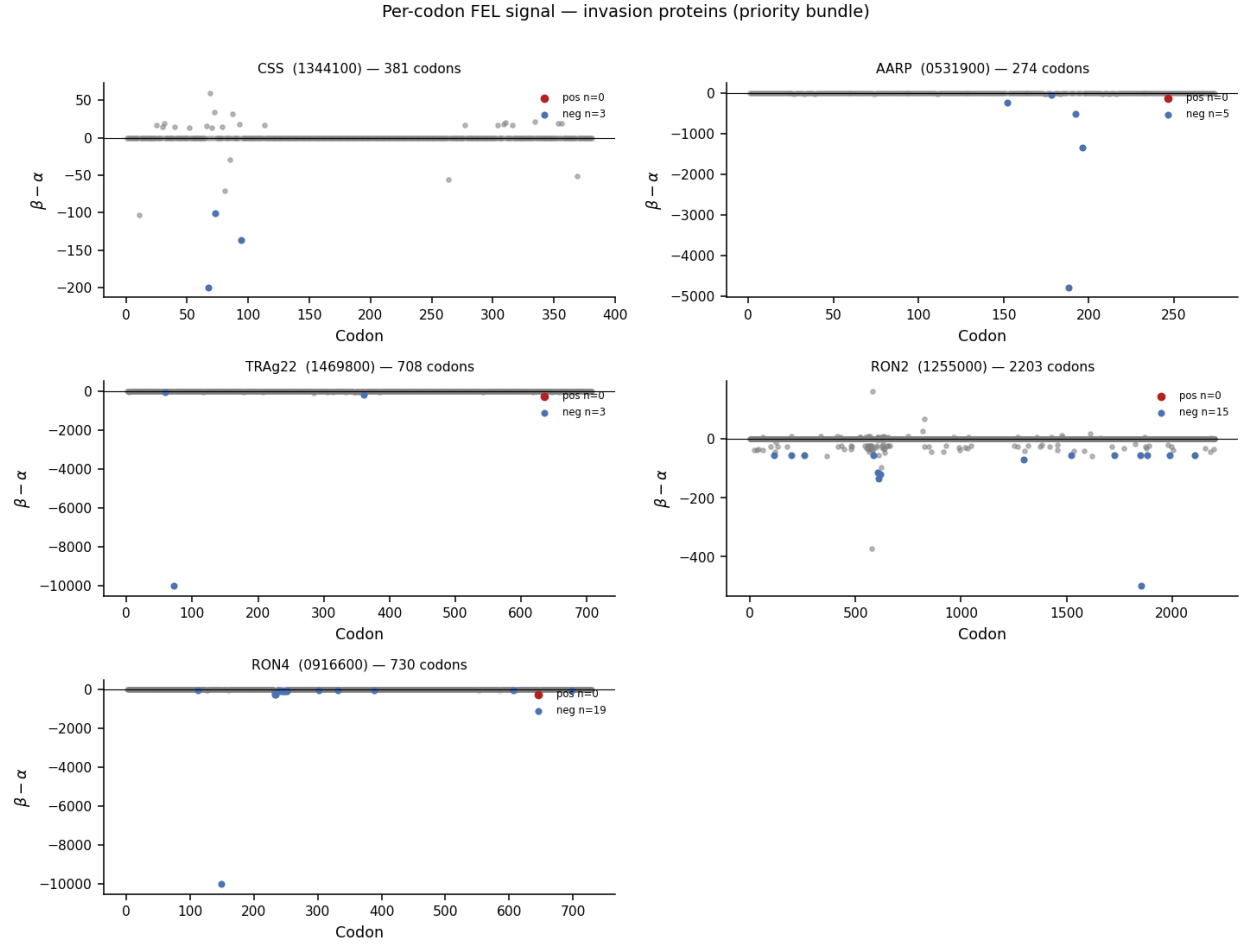


Figure 7: Per-codon FEL signal for invasion / erythrocyte-binding priority genes

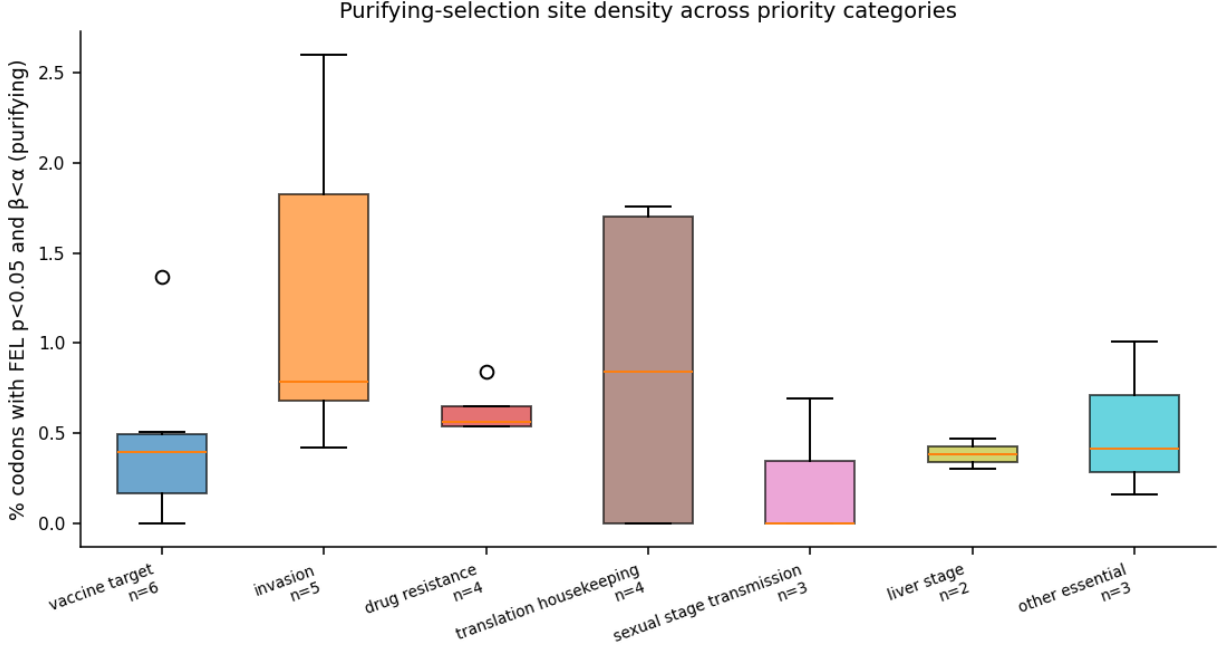


Figure 8: Purifying-site density by priority category

2. **Relaxed Phase H rebuild** at min_intact=5 (currently 7) to recover the 116 missing priority genes—the rebuild trades 2 strains of breadth for the full vaccine / drug-resistance / liver-stage panel.
3. **HyPhy RELAX** between Sal-I (lab-adapted) and the 7 field/clinical strains to test for relaxation vs intensification of selection on the lab line.
4. **McDonald-Kreitman** with the MalariaGEN *P. vivax* genomic-epidemiology VCFs—within-species polymorphism vs cross-strain divergence—for the high-priority hits that survive recommendations 1-2.

6. Tables

6.1 Top 50 bulk BUSTED hits

Rank	Gene	PVP01 id	Chr	Family	Category	p (BUSTED)	-log10(p)
1	—	PVP01_1255400	chr12	other	non-priority	0.00e+00	300.00
2	—	PVP01_0815500	chr8	other	non-priority	1.18e-09	8.93
3	—	PVP01_0929400	chr9	other	non-priority	3.04e-07	6.52
4	—	PVP01_0104700	chr1	other	non-priority	1.86e-06	5.73
5	—	PVP01_1027200	chr10	other	non-priority	3.89e-06	5.41
6	—	PVP01_1219200	chr12	hypothetical	non-priority	1.12e-05	4.95
7	—	PVP01_1110200	chr11	other	non-priority	2.89e-05	4.54
8	—	PVP01_1220100	chr12	hypothetical	non-priority	2.92e-05	4.53
9	—	PVP01_1135000	chr11	other	non-priority	3.22e-05	4.49
10	—	PVP01_0306600	chr3	other	non-priority	4.06e-05	4.39
11	—	PVP01_1258300	chr12	other	non-priority	6.89e-05	4.16
12	—	PVP01_1227100	chr12	other	non-priority	7.31e-05	4.14
13	—	PVP01_1334300	chr13	other	non-priority	1.08e-04	3.97
14	—	PVP01_1120000	chr11	other	non-priority	1.98e-04	3.70
15	—	PVP01_1323000	chr13	other	non-priority	2.26e-04	3.65
16	—	PVP01_0109200	chr1	other	non-priority	2.64e-04	3.58

Rank	Gene	PVP01 id	Chr	Family	Category	p (BUSTED)	-log10(p)
17	CSS	PVP01_1344100	chr13	other	invasion	3.92e-04	3.41
18	—	PVP01_0115400	chr1	other	non-priority	4.52e-04	3.34
19	—	PVP01_0909900	chr9	conserved (housekeeping-like)	non-priority	8.14e-04	3.09
20	—	PVP01_0715700	chr7	other	non-priority	9.04e-04	3.04
21	RON2	PVP01_1255000	chr12	other	invasion	1.13e-03	2.95
22	—	PVP01_0107900	chr1	other	non-priority	1.51e-03	2.82
23	—	PVP01_1252800	chr12	conserved (housekeeping-like)	non-priority	2.06e-03	2.69
24	—	PVP01_0616400	chr6	other	non-priority	2.35e-03	2.63
25	—	PVP01_0409900	chr4	other	non-priority	2.85e-03	2.54
26	—	PVP01_1427700	chr14	other	non-priority	3.01e-03	2.52
27	AARP	PVP01_0531900	chr5	other	invasion	3.13e-03	2.50
28	—	PVP01_1464900	chr14	other	non-priority	3.16e-03	2.50
29	tRNA	PVP01_0815300	chr8	non-coding	non-priority	3.30e-03	2.48
30	—	PVP01_0913000	chr9	other	non-priority	3.70e-03	2.43
31	—	PVP01_0926500	chr9	other	non-priority	3.83e-03	2.42
32	—	PVP01_0944000	chr9	other	non-priority	4.24e-03	2.37
33	—	PVP01_1406800	chr14	conserved (housekeeping-like)	non-priority	4.66e-03	2.33
34	—	PVP01_1341500	chr13	conserved (housekeeping-like)	non-priority	4.71e-03	2.33
35	—	PVP01_0707500	chr7	conserved (housekeeping-like)	non-priority	4.89e-03	2.31
36	—	PVP01_1455000	chr14	conserved (housekeeping-like)	non-priority	5.77e-03	2.24
37	—	PVP01_1107700	chr11	other	non-priority	5.81e-03	2.24
38	—	PVP01_1243600	chr12	other	non-priority	5.85e-03	2.23
39	—	PVP01_0815600	chr8	other	non-priority	5.96e-03	2.22
40	—	PVP01_1469500	chr14	hypothetical	non-priority	6.21e-03	2.21
41	—	PVP01_1444900	chr14	other	non-priority	6.90e-03	2.16
42	—	PVP01_1133300	chr11	other	non-priority	6.92e-03	2.16
43	—	PVP01_1267500	chr12	other	non-priority	7.05e-03	2.15
44	—	PVP01_0214500	chr2	conserved (housekeeping-like)	non-priority	7.53e-03	2.12
45	—	PVP01_0213000	chr2	other	non-priority	9.19e-03	2.04
46	—	PVP01_1320300	chr13	other	non-priority	9.30e-03	2.03
47	—	PVP01_1121800	chr11	other	non-priority	9.37e-03	2.03
48	—	PVP01_1142800	chr11	other	non-priority	9.93e-03	2.00
49	rRNA	PVP01_1117500	chr11	non-coding	non-priority	9.94e-03	2.00
50	—	PVP01_1262300	chr12	other	non-priority	1.03e-02	1.99

(Full top-200 table at [writeup/top200_bulk_busted.tsv](#).)

6.2 Priority bundle — all 41 genes, four methods

Gene	PVP01 id	Category	BUSTED p	aBSREL min p	MEME min p	FEL+	FEL−	n sites	n branches sig (aB- SREL)
SIAP1	PVP01_0307900	ribosome_stage	5.00e-01	1.00e+00	5.25e-02	0	3	997	0/10
Pvs230	PVP01_0415800	ribosome_target	5.59e-02	2.69e-02	1.38e-03	1	9	2729	1/13
HSP70-1	PVP01_0515400	translation_housekeeping	5.30e-01	1.00e+00	6.67e-01	0	12	682	0/8
AARP	PVP01_0531900	ribosome	1.68e-02	3.78e-06	1.81e-02	0	5	274	1/12
Pvs25	PVP01_0616400	ribosome_target	5.00e-01	1.00e+00	1.44e-01	0	3	219	0/12
PVP01_0808600	PVP01_0808600	ribosome	5.00e-01	1.00e+00	6.67e-01	0	1	250	0/8
HAP2	PVP01_0814800	ribosome_target	1.60e-01	1.00e+00	3.13e-02	0	1	860	0/10
Enolase	PVP01_0816000	metabolism	5.00e-01	1.00e+00	6.67e-01	0	3	446	0/9
Vivapain-2	PVP01_0916000	ribosome_essential	5.00e-01	1.00e+00	2.15e-01	0	2	484	0/8
Vivapain-3	PVP01_0916400	ribosome_essential	1.64e-01	1.00e+00	1.15e-01	0	5	495	0/11
PVP01_0916600	PVP01_0916600	ribosome	5.00e-01	1.00e+00	4.37e-02	0	6	487	0/9
RON4	PVP01_0916600	ribosome	1.06e-01	1.00e+00	8.91e-02	0	19	730	0/11
AMA1	PVP01_0934200	ribosome_target	3.90e-01	1.53e-03	3.60e-02	0	0	562	2/13
PVM-exported	PVP01_0934800	cell_adherence	5.00e-01	1.00e+00	3.57e-01	0	2	663	0/8
HSP									
TCTP	PVP01_1023000	drug_resistance	5.00e-01	1.00e+00	6.67e-01	0	1	171	0/9
PVP01_1117400	PVP01_1117400	ribosome	5.00e-01	1.00e+00	6.67e-01	0	0	181	0/8
Hexokinase	PVP01_1125500	metabolism	5.00e-01	1.00e+00	4.19e-01	0	4	493	0/8
Histone H2A	PVP01_1131700	translation_housekeeping	5.30e-01	1.00e+00	6.67e-01	0	0	133	0/8
DHODHP	PVP01_1145600	drug_resistance	5.00e-01	1.00e+00	2.57e-01	0	3	557	0/10
Pvs47	PVP01_1208000	ribosome_target	7.59e-02	1.00e+00	3.15e-02	0	2	433	0/13
K12	PVP01_1211100	drug_resistance	5.00e-01	1.00e+00	3.51e-01	0	6	712	0/8
PEPCK	PVP01_1212000	metabolism	5.00e-01	1.00e+00	3.25e-01	0	2	599	0/8
PVP01_1216600	PVP01_1216600	ribosome	5.00e-01	1.00e+00	4.28e-01	0	6	513	0/10
SIR2A	PVP01_1225700	histone_matin_regulation	5.00e-01	1.00e+00	3.71e-01	0	2	306	0/8
G6PD	PVP01_1253200	drug_resistance	5.00e-01	1.00e+00	4.33e-02	0	5	927	0/9
RON2	PVP01_1255000	ribosome	1.12e-03	5.63e-06	1.34e-02	0	15	2203	1/12
EF-2	PVP01_1255000	translation_housekeeping	5.30e-01	1.00e+00	6.67e-01	0	14	832	0/9
GEST	PVP01_1258000	ribosome_target	5.00e-01	1.00e+00	7.50e-02	0	0	249	0/9
M17	PVP01_1260800	ribosome_essential	5.00e-01	1.00e+00	5.69e-02	0	1	621	0/8
aminopeptidase									
Aldolase	PVP01_1262200	metabolism	5.00e-01	1.00e+00	6.67e-01	0	5	369	0/9
Actin-II	PVP01_1336400	translation_housekeeping	5.30e-01	1.00e+00	6.67e-01	0	0	376	0/8
CSS	PVP01_1344100	ribosome	3.92e-04	2.91e-04	1.41e-01	0	3	381	1/12
GAP	PVP01_1403800	ribosome_target	5.00e-01	1.00e+00	1.32e-01	0	0	244	0/11
UIS3	PVP01_1403100	ribosome_target	5.00e-01	1.00e+00	6.67e-01	0	1	213	0/8
Ferredoxin	PVP01_1419000	chloroplast_markers	2.19e-01	1.00e+00	1.80e-01	0	1	196	0/8
MGET	PVP01_1435800	ribosome_target	5.00e-01	1.00e+00	1.20e-01	0	2	288	0/9
CelTOS	PVP01_1435400	ribosome_target	5.00e-01	1.00e+00	5.17e-02	0	1	196	0/9

Gene	PVP01 id	Category	BUSTED p	aBSREL min p	MEME min p	FEL+	FEL−	n sites	n branches sig (aB- SREL)
HP1	PVP01_1439200	translamin_regul	5.00e-01	1.00e+00	6.67e-01	0	0	275	0/8
ASF1	PVP01_1442700	translamin_regul	3.99e-01	1.00e+00	2.02e-01	0	2	269	0/9
PVP01_P46000	PVP01_1460500	der	5.00e-01	1.00e+00	6.67e-01	0	0	115	0/8
TRAg22	PVP01_1469800	asion	2.39e-01	2.29e-01	8.61e-02	0	3	708	0/11

(Full priority bundle table at `writeup/priority_bundle_results.tsv`.)

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